

NuMicro[®] Family VS Code Quick Start Guide

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TABLE OF CONTENTS

1 INSTALLATION VS CODE AND EXTENSIONS 3

2 GET STARTED WITH A SAMPLE PROJECT 4

3 MANAGE ARM LICENSE 7

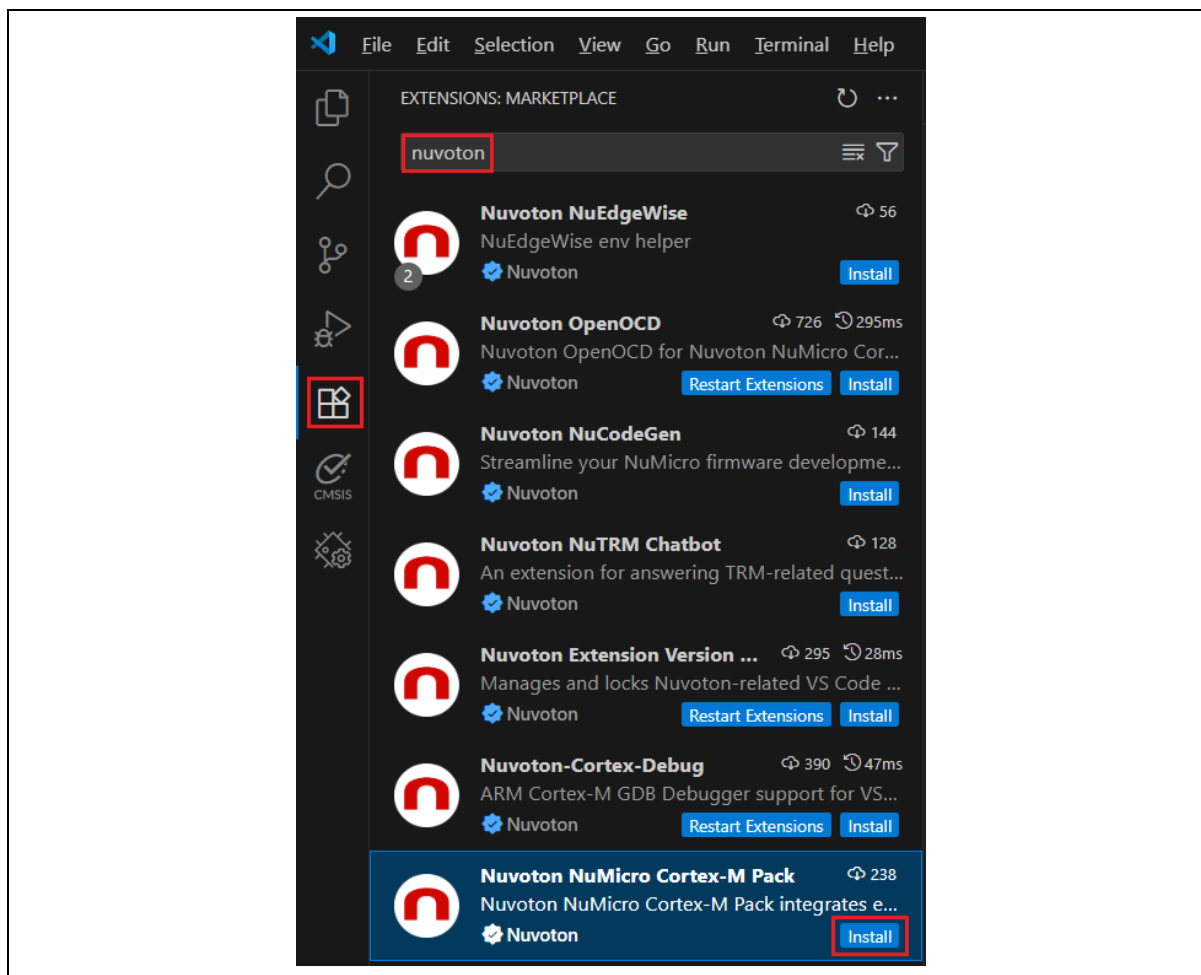
4 CONFIGURE THE DEVICE 9

5 RUN THE SAMPLE PROJECT 12

6 ADDITIONAL NOTE 15

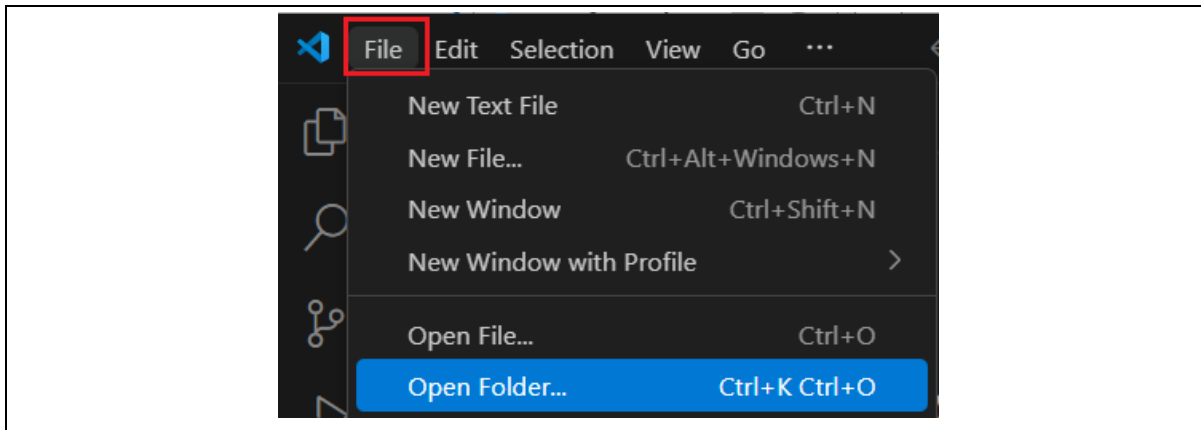
1 INSTALLATION VS CODE AND EXTENSIONS

1. Download VS Code from <https://code.visualstudio.com> and install it.
2. Launch VS Code and click Extensions in the Activity Bar.
3. Text “**Nuvoton NuMicro Cortex-M Pack Extension**” in search bar. Click the “**Install**” button to install.

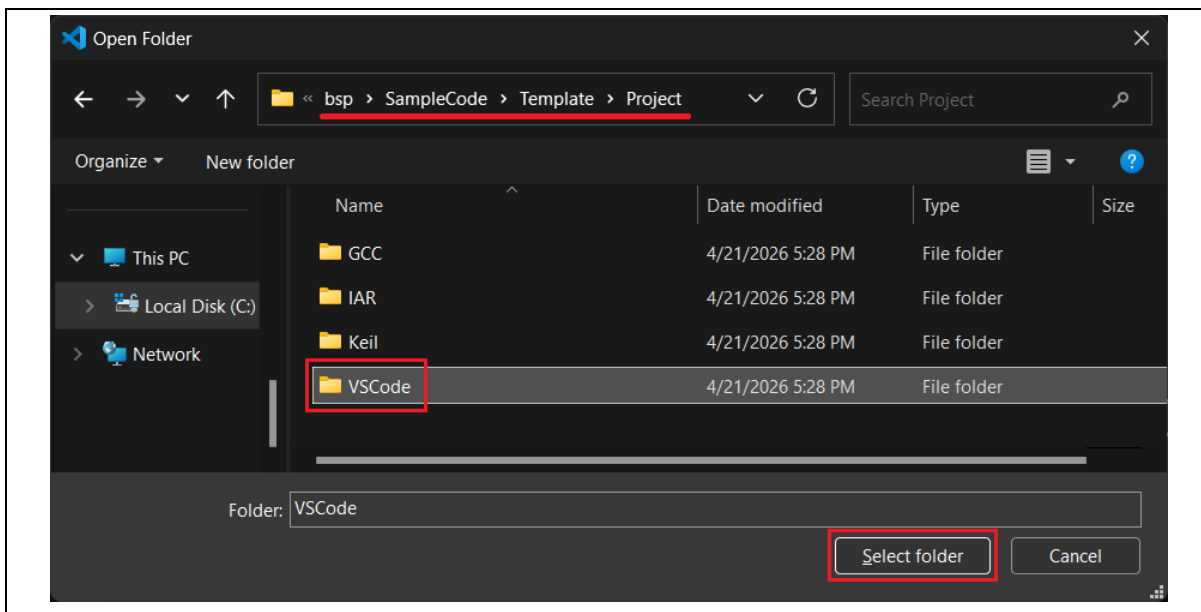


2 GET STARTED WITH A SAMPLE PROJECT

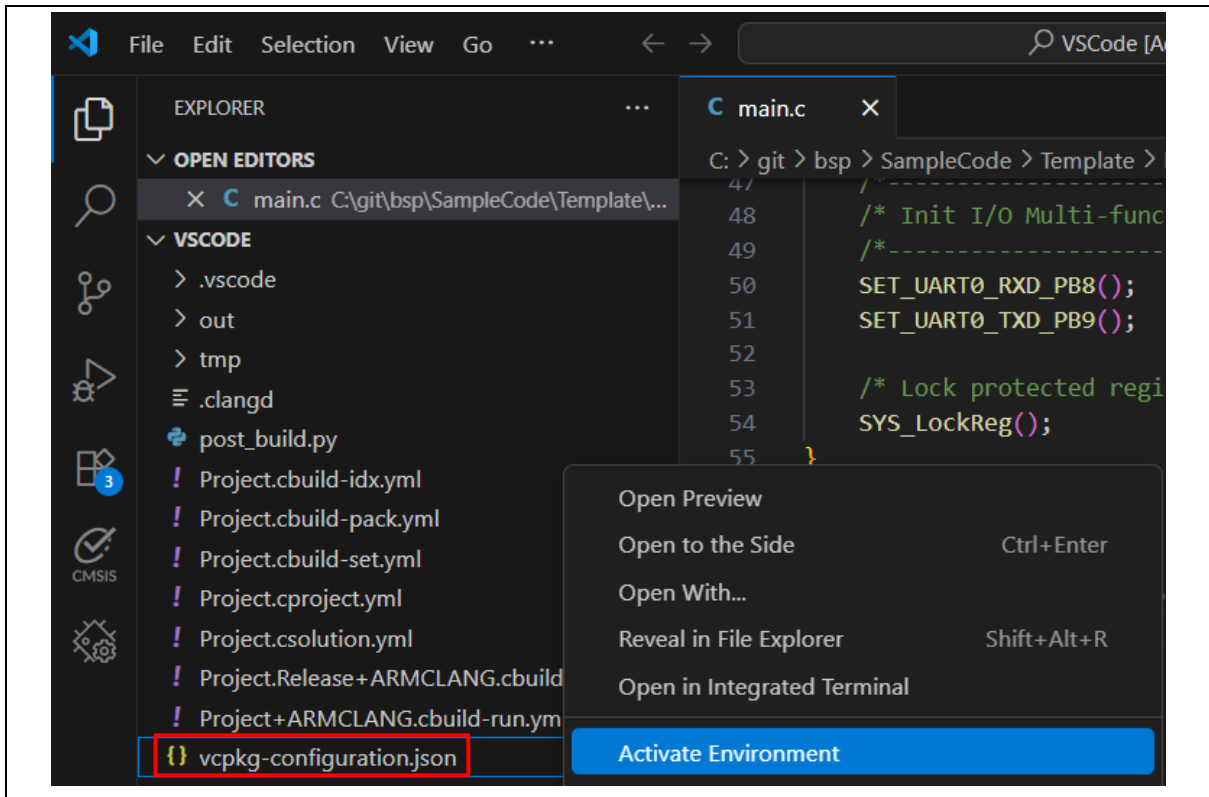
1. Click "File" menu and select "Open Folder" in the toolbar.



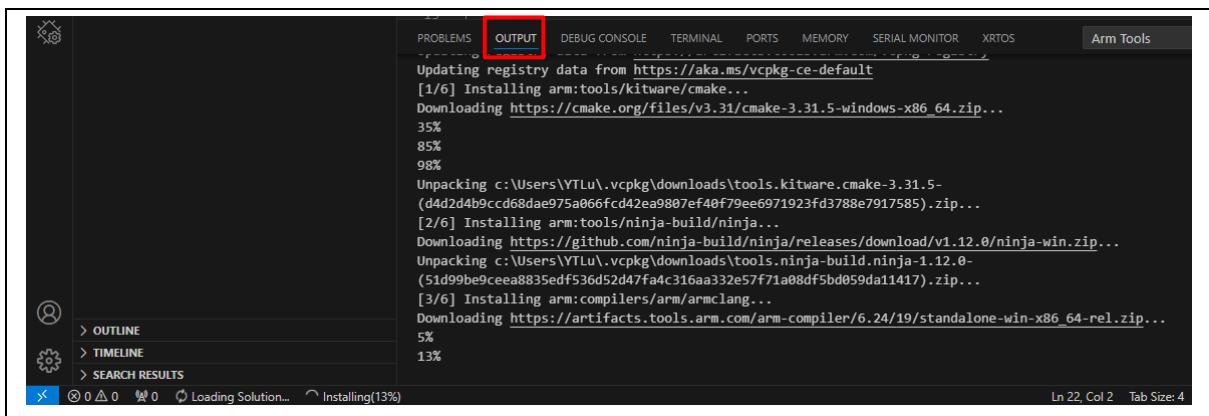
2. Select the "VSCode" folder of a sample project to open the project.



- Right-click “**vcpkg-configuration.json**” file and select “**Activate Environment**” command. (Please use private network connection.)

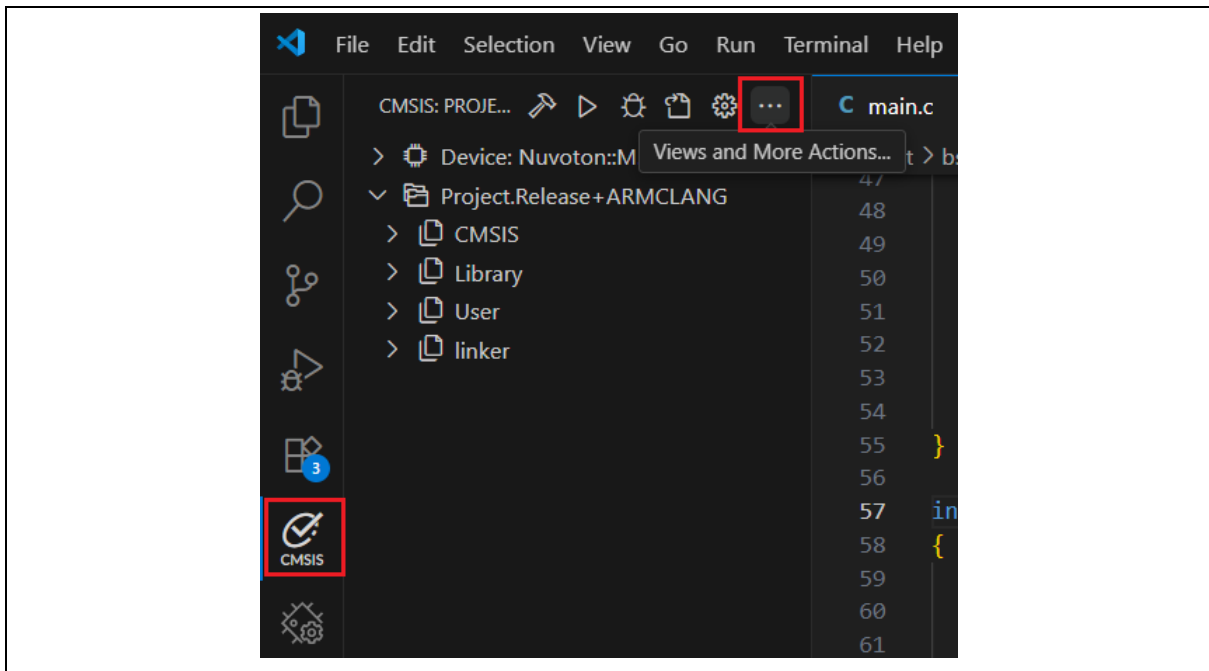


- Check **OUTPUT** terminal at the bottom. It will download and install the requires tools.

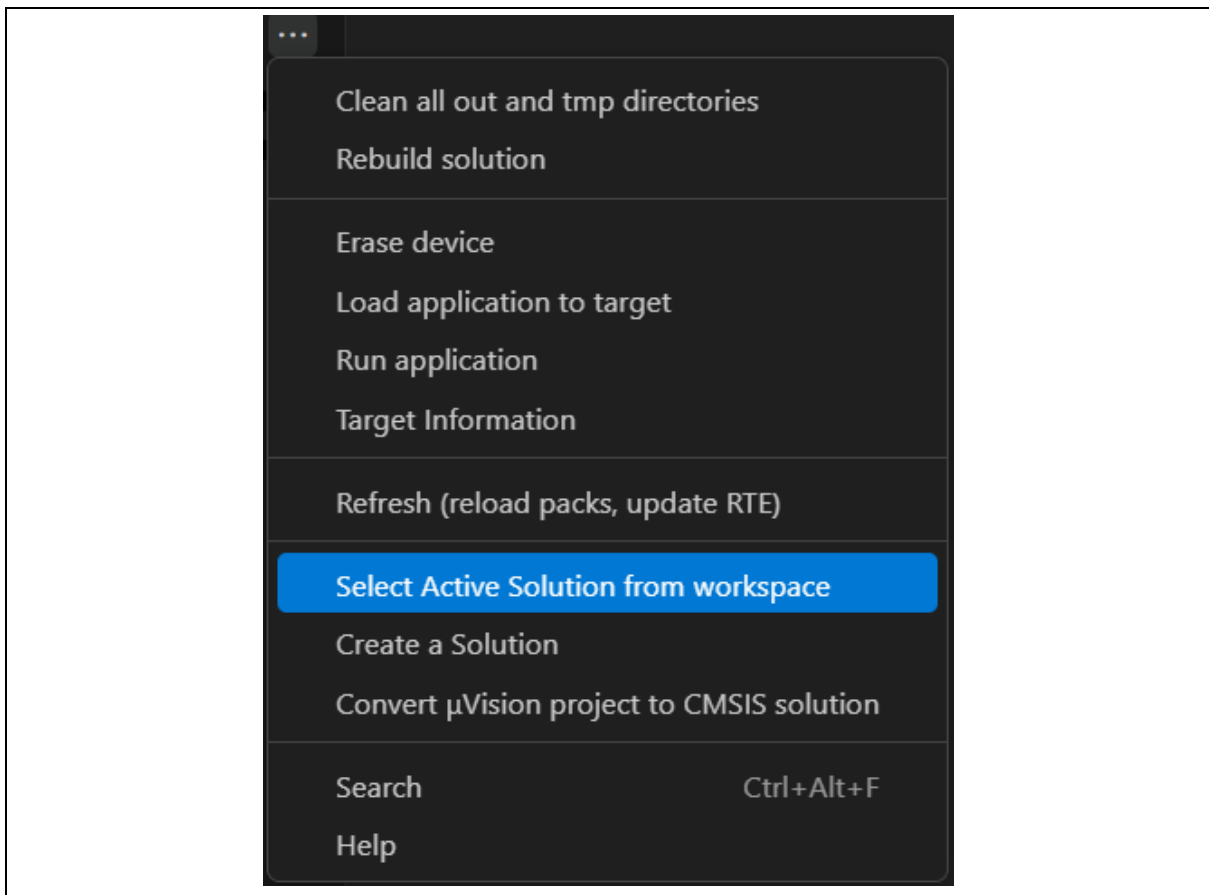


5. Click **CMSIS** icon in the Activity Bar.

Then click **“Views and More Commands”** button.

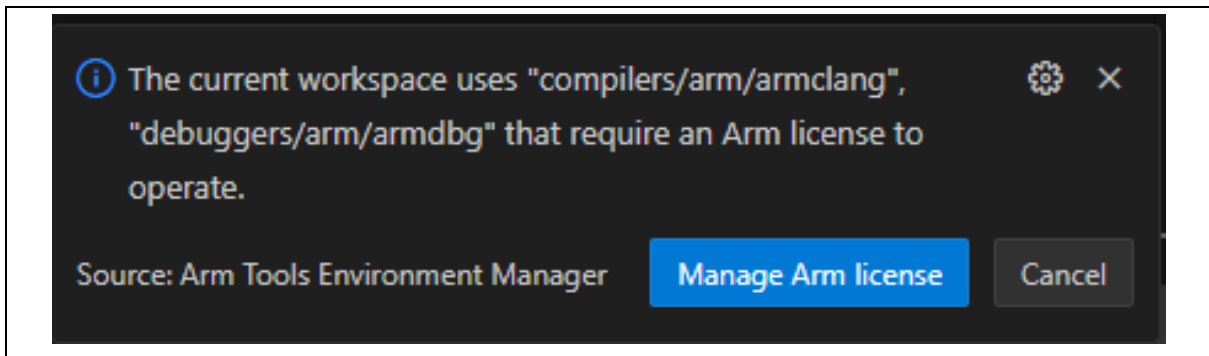


6. Select **“Select Active solution from workspace”** command.

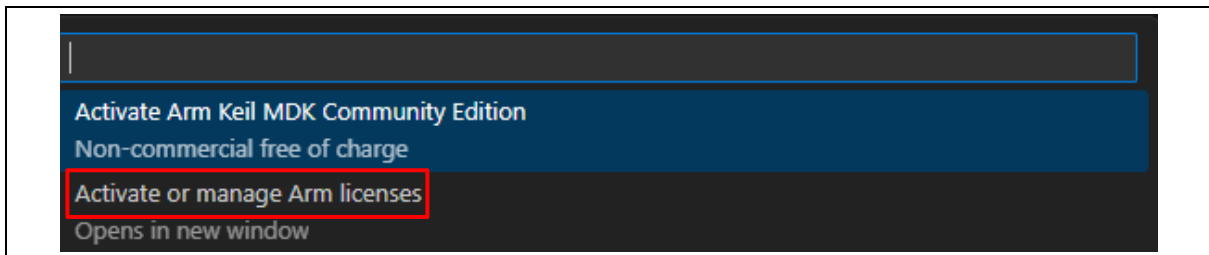


3 MANAGE ARM LICENSE

1. Click **"Manage Arm License"** button in the pop-up window at the bottom right.

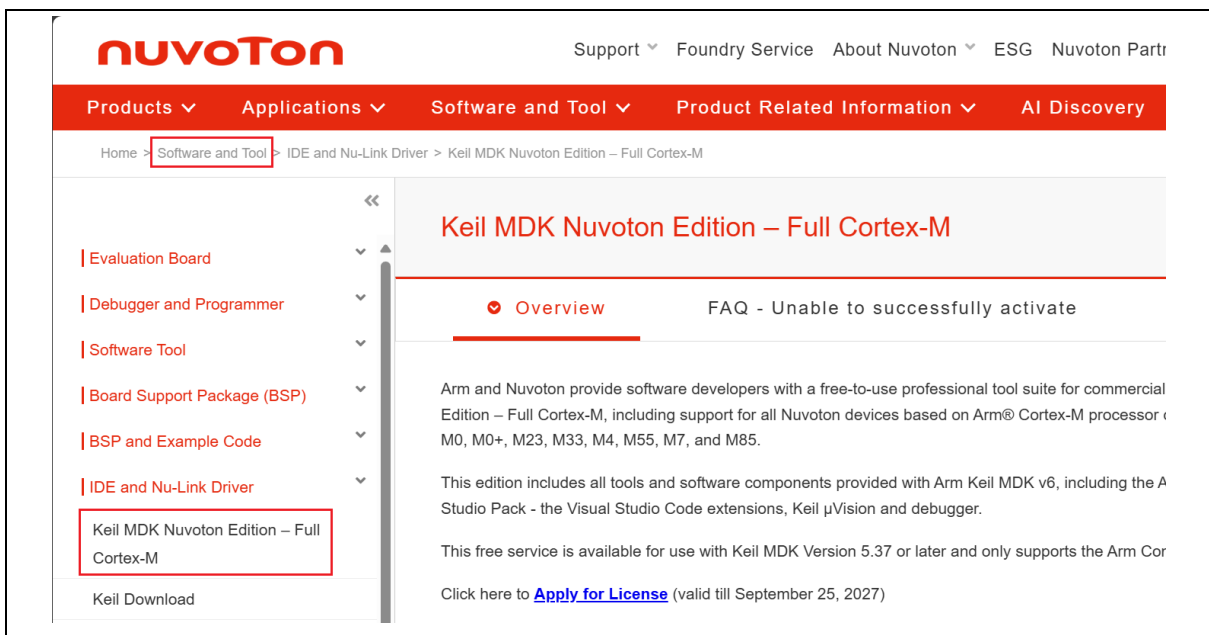


2. In the search bar at the top, select **"Activate or manage Arm licenses"** command from the available options.



3. Get Keil MDK License ID code.

- 1) Navigation to [Official Website](#).



2) Fill out the form.

Apply for Keil MDK Nuvoton Edition – Full Cortex-M

First Name*	Email*
Last Name*	Phone*
Company / Organization *	Industry
	-- Industry --
Job Title *	Application*
	-- Application --
Region / Country or region *	Series*
-- Select Region --	-- Select Line --
State / Province*	-- Select Category --
	Part No

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 NUVOTON Privacy Policy
 Last modified: June 01, 2023

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Submit

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3) Check mailbox and fill in the License Server.

Arm License Management Utility

Activate your product

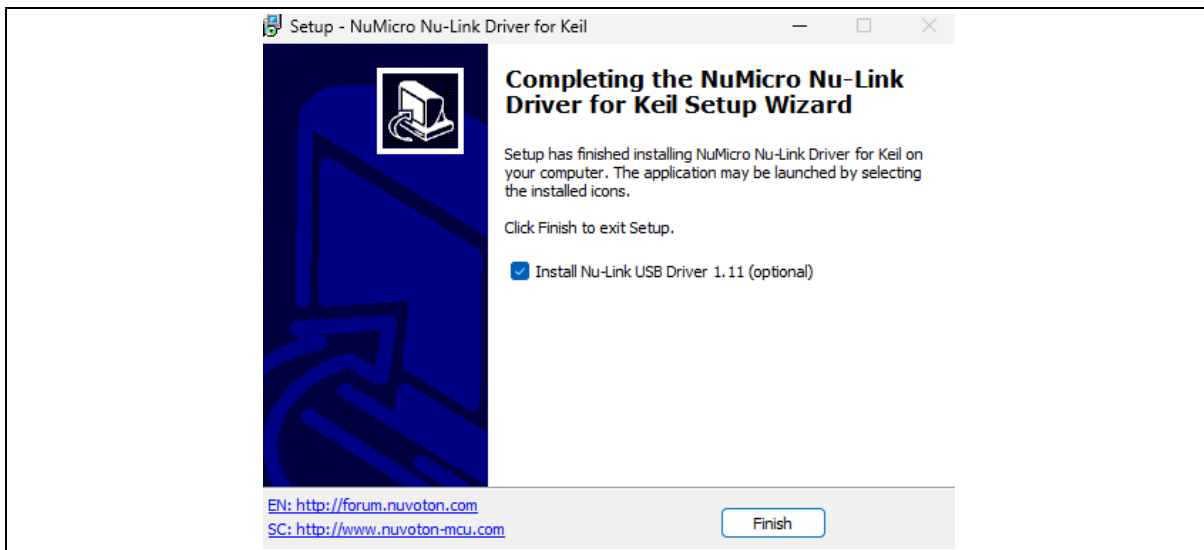
Activate with: ☐ Activation Code ☒ License Server

Enter your license server address:

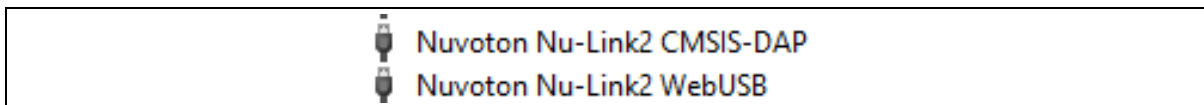
Query

4 CONFIGURE THE DEVICE

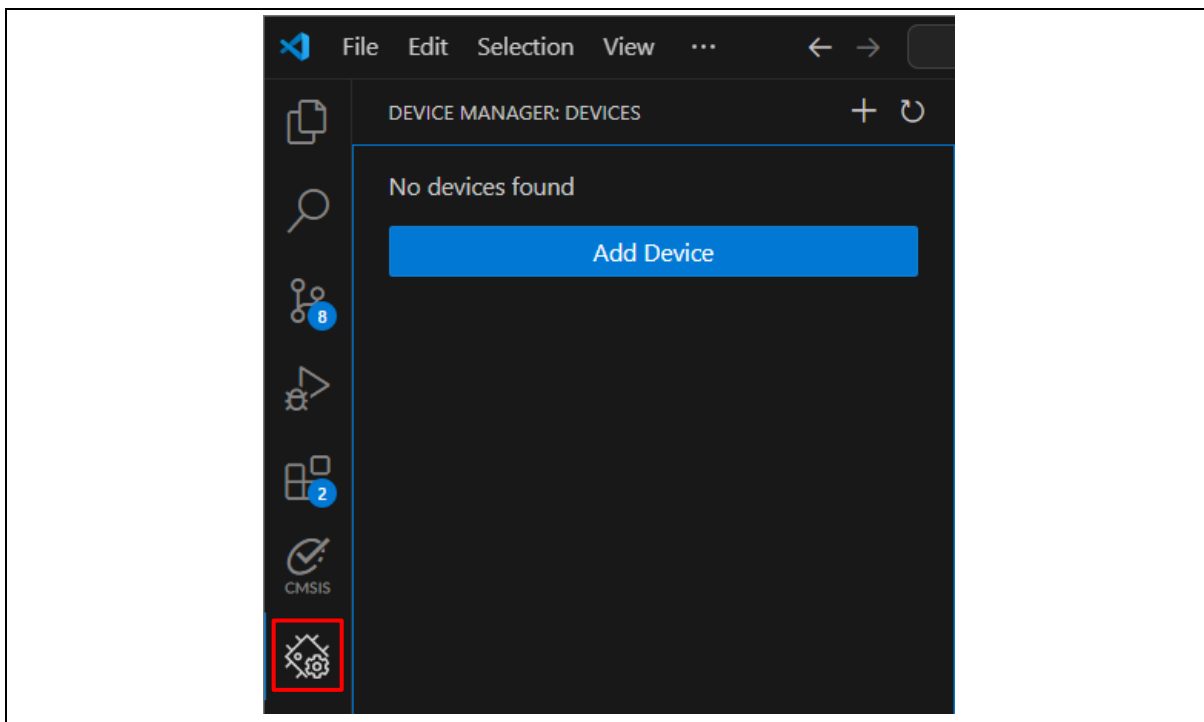
1. Install Nuvoton Nu-Link Keil Driver.
2. After installing the Keil driver, please check the box to install the Nu-Link USB Driver.



3. Installation complete in Windows Device Manager.



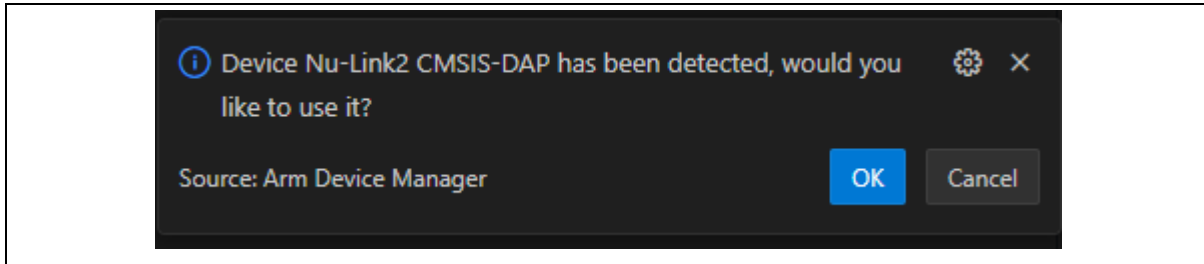
4. Click “**Device Manager**” icon in the Activity Bar to open the Device Manager.



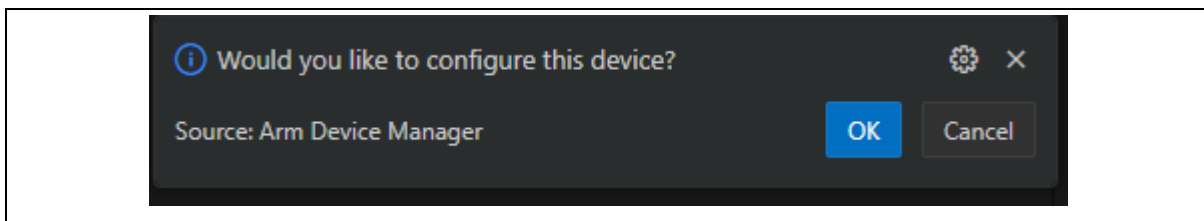
5. Connect device to your computer over USB.

The Device Manager detects the board and displays a pop-up message.

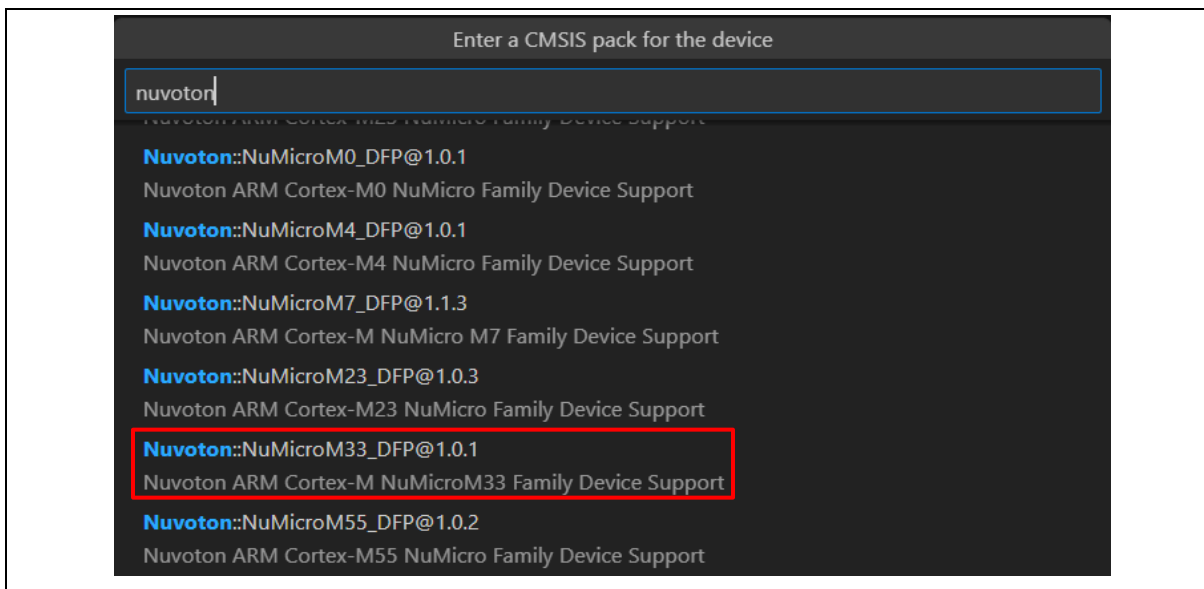
Press OK in the pop-up message and use it.



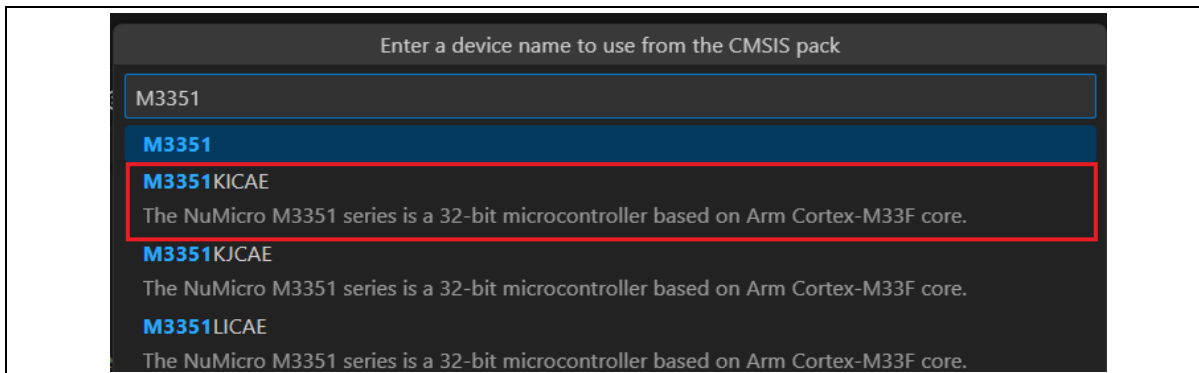
6. Press OK in the pop-up message and configure this device.



7. Text “**nuvoton**” in search bar and select corresponding CMSIS pack for the device.

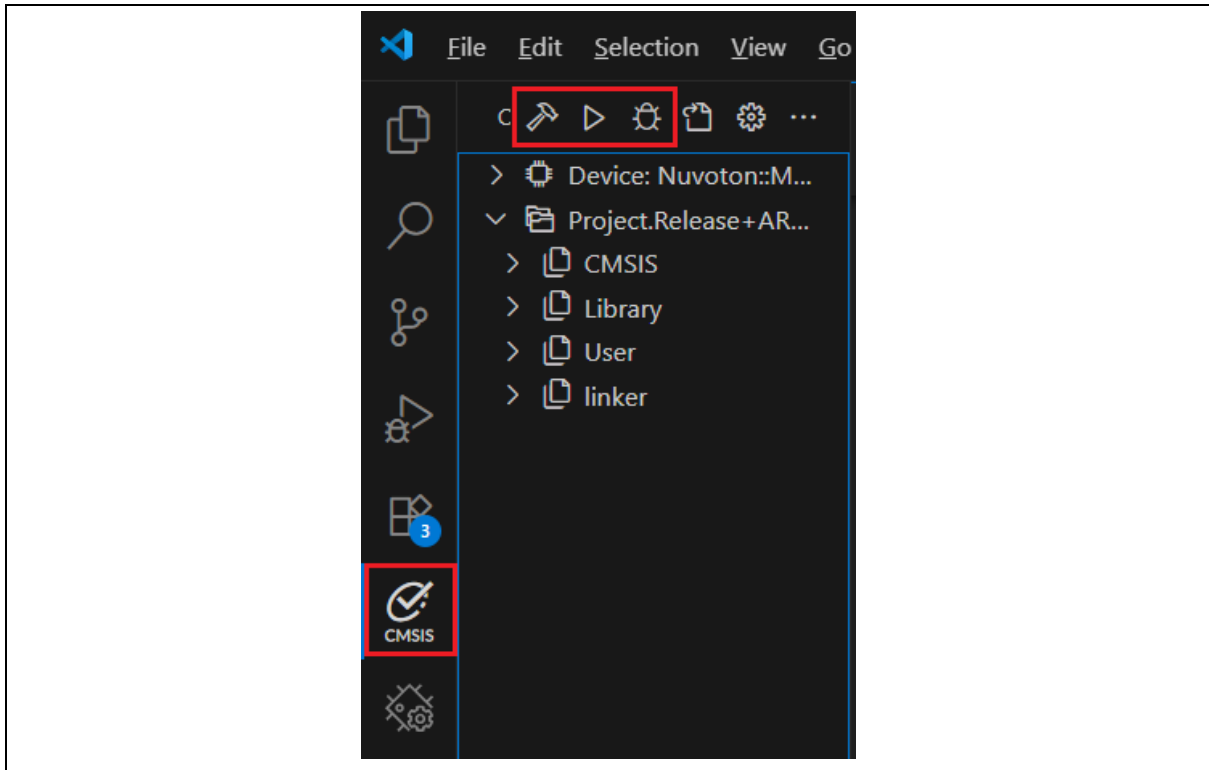


8. Text your “**device model number**” in search bar.

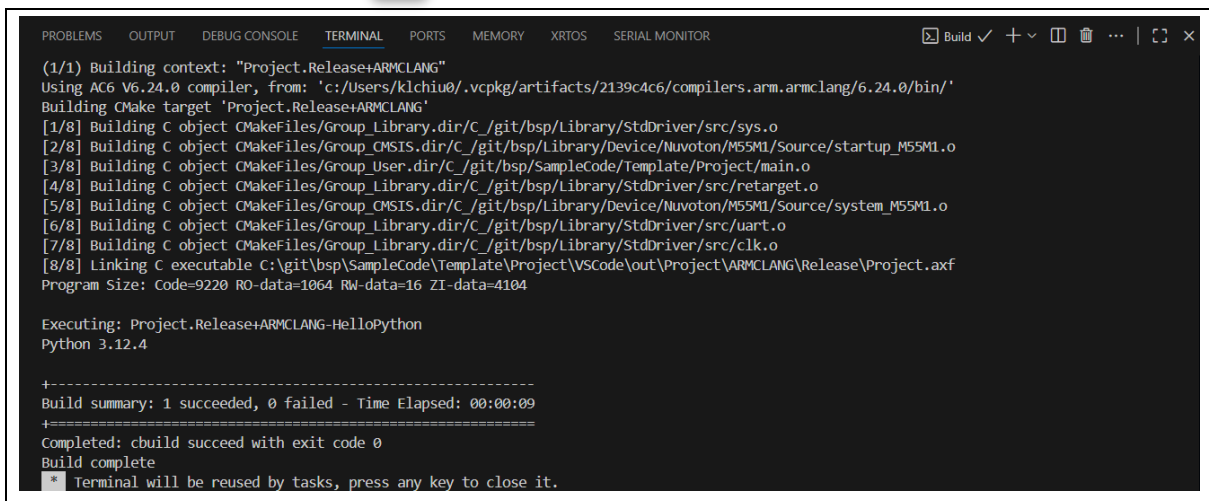


5 RUN THE SAMPLE PROJECT

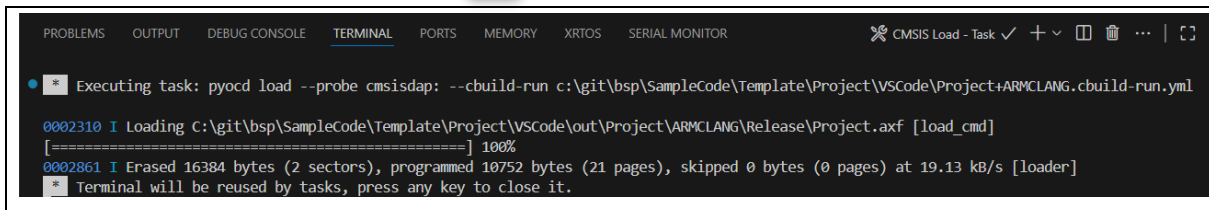
1. Click “CMSIS” icon in the Activity Bar.



2. Click “Build” button.

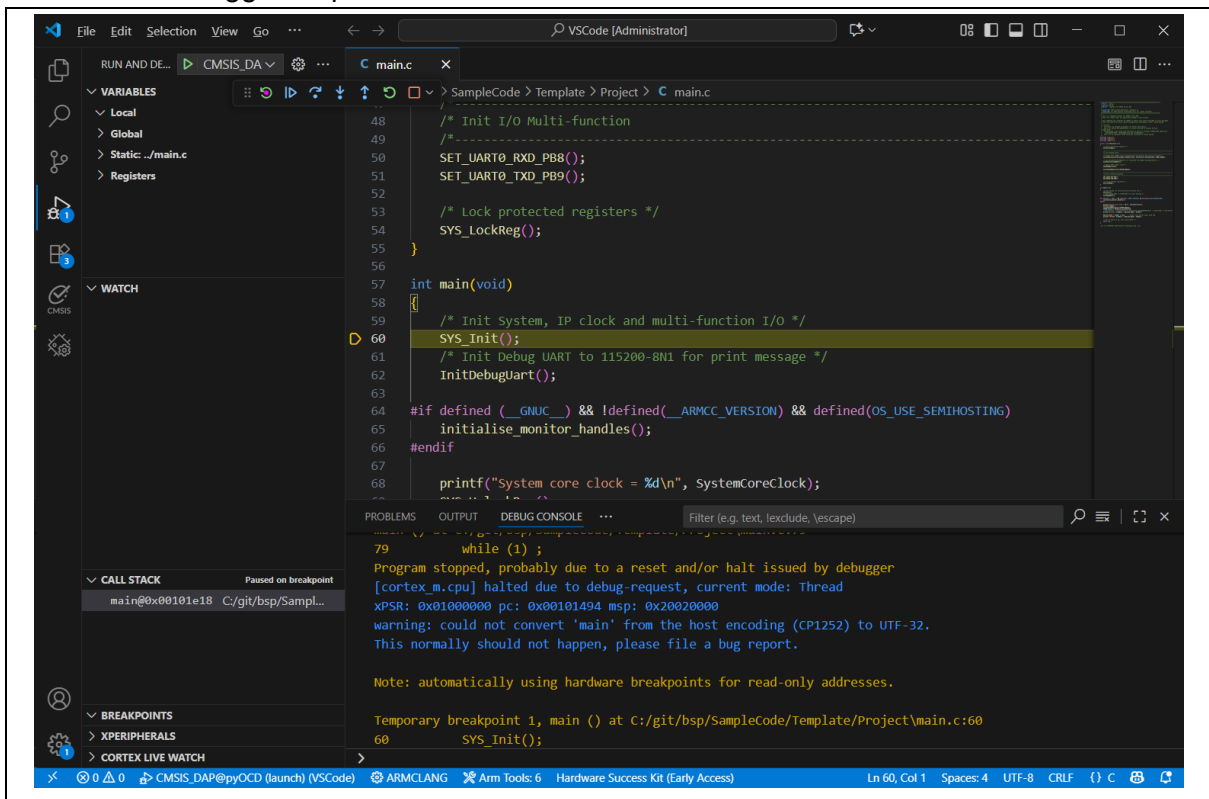


3. Click “Load & Run” button. 

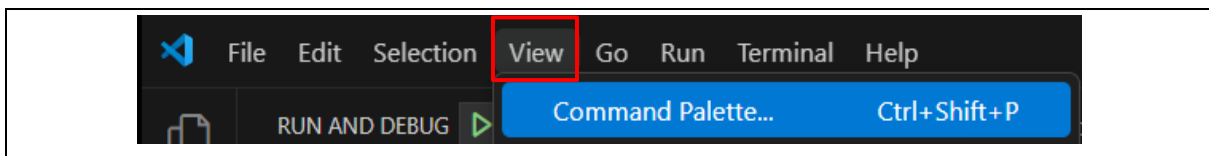


4. Click “Load & Debug” button. 

The debugger stops at the main function.



5. Click “View” menu and select “Command palette” item in the toolbar.



The screenshot displays the Keil IDE interface. The main window shows the source code for 'main.c', which includes a system clock definition of 30MHz and a UART module initialization. The 'PinView' window on the right shows the pin configuration for the M3351KJCAE, with pins 97 through 154 assigned to various functions like UART_TXD, UART_RXD, and I2C_SDA.

Source Code (main.c):

```

1  #include "main.h"
2
3  /* Enable UART module clock */
4  SetDebugClock();
5
6  /*
7   *-----
8   * Init I/O Multi-function
9   *-----
10 */
11 SetDebugIOP();
12
13 /* Lock protected registers */
14 Sys_Locking();
15
16 return 0;
17
18 }
19
20 int main(void)
21 {
22     /* Init System, IP clock and multi-function I/O */
23     Sys_Init();
24     /* Init debug UART to 115200 Bps for print message */
25     InitDebugIOP();
26
27     #if defined(_DRC_) && !defined(_AMEC_VERSION) && defined(ON_USD_SMPD)
28     Initialise_monitor_handles();
29     #endif
30
31     printf("System core clock: %dHz", SystemCoreClock);
32     printf("Hello World!\n");
33
34     /* Infinite loop to prevent from exiting */
35     while (1)
36     {
37         // Add custom routine here.
38     }
39
40     /* (c) COPYRIGHT 2025 Nuvoton Technology Corp. *** */
41 }

```

PinView Window:

PinView - PinView X

PinView shows the pin configuration for the M3351KJCAE. The pins are listed in a table with their functions and pin numbers.

Pin	Function
97	UART_TXD
98	UART_RXD
99	UART_TXD
100	UART_RXD
101	UART_TXD
102	UART_RXD
103	UART_TXD
104	UART_RXD
105	UART_TXD
106	UART_RXD
107	UART_TXD
108	UART_RXD
109	UART_TXD
110	UART_RXD
111	UART_TXD
112	UART_RXD
113	UART_TXD
114	UART_RXD
115	UART_TXD
116	UART_RXD
117	UART_TXD
118	UART_RXD
119	UART_TXD
120	UART_RXD
121	UART_TXD
122	UART_RXD
123	UART_TXD
124	UART_RXD
125	UART_TXD
126	UART_RXD
127	UART_TXD
128	UART_RXD
129	UART_TXD
130	UART_RXD
131	UART_TXD
132	UART_RXD
133	UART_TXD
134	UART_RXD
135	UART_TXD
136	UART_RXD
137	UART_TXD
138	UART_RXD
139	UART_TXD
140	UART_RXD
141	UART_TXD
142	UART_RXD
143	UART_TXD
144	UART_RXD
145	UART_TXD
146	UART_RXD
147	UART_TXD
148	UART_RXD
149	UART_TXD
150	UART_RXD
151	UART_TXD
152	UART_RXD
153	UART_TXD
154	UART_RXD

The PinView window also shows the pin configuration for the M3351KJCAE, with pins 97 through 154 assigned to various functions like UART_TXD, UART_RXD, and I2C_SDA.

6 ADDITIONAL NOTE

1. How to Fix Windows Long Path Errors in VS Code

If you encounter the following compilation errors while working in VS Code, your project likely exceeds the default Windows path limit:

- 1) The maximum full path to an object file is 250 characters.

```
ase/CMakeFiles/CMakeScratch/TryCompile-n0omuq/CMakeFiles/cmTC_a82c8.dir/./
has 251 characters. The maximum full path to an object file is 250
characters (see CMAKE_OBJECT_PATH_MAX). Object file
CMakeCCompilerABI.o
cannot be safely placed under this directory. The build may not work
correctly.
```

- 2) The system cannot find the path specified.

```
LANG-stamp/Template.Release+ARMCLANG-configure-.cmake:49 (message):
Command failed: The system cannot find the path specified
```

To resolve this, follow these steps to enable Long Paths:

- 1) Open Registry Editor: Press Win + R, type regedit, and press Enter.
- 2) Navigate to: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem
- 3) Enable Feature: Double-click LongPathsEnabled and change the Value data from 0 to 1.
- 4) Apply: Click OK and restart your computer (or restart VS Code) to apply the changes.

For more technical details on why this limit exists and how Windows handles it, please refer to the official Microsoft documentation:

[Microsoft Learn: Maximum Path Length Limitation](#)

REVISION HISTORY

Date	Revision	Description
2026.02.12	1.02	1. Initial version.

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